Easy To Forget: A Data Selection Strategy To Mitigate Catastrophic Forgetting

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Abstract

Continual learning is vital for intelligent systems, enabling them to adapt to dynamic environments without forgetting previously acquired knowledge. However, deep neural networks suffer from catastrophic forgetting, which challenges their ability to retain past information when exposed to new data. Experience Replay (ER) is a promising technique to mitigate this issue by storing and rehearsing subsets of prior data during training. This project presents a novel data selection strategy for ER, focusing on mitigating catastrophic forgetting in continual learning. Our proposed approach, called EF (Easy-to-Forget), operates under the assumption that "easy-to-forget" data is also "hard-to-learn." Accordingly, we emphasize misclassified data, making it more likely to be added to the buffer. Using CIFAR-10, CIFAR-100, and MNIST datasets, our experiments assess classification accuracy and forgetting rate, shedding light on optimal strategies for robust continual learning.

1. Introduction

Continual learning has emerged as a critical research area for enabling intelligent systems to adapt to ever-changing environments (Wang et al., 2024). By incrementally integrating new information, these systems preserve previously learned knowledge while incorporating fresh insights. Real-world settings—such as finance, robotics, and personalized services—often involve continuous, non-stationary data streams, underscoring the need for models that can adapt in real time (Verwimp et al., 2023). However, conventional training approaches, typically developed for static datasets, struggle with dynamic input distributions (Gunasekara et al., 2023).

Despite its promise, continual learning for deep neural networks often suffers from *catastrophic forgetting*, where performance on previously learned tasks deteriorates after training on new data (McCloskey & Cohen, 1989; Ratcliff, 1990; French, 1999; Goodfellow et al., 2013). This is mainly because standard training updates all parameters simultaneously, overwriting prior representations. In many real-world situations, preserving or extending model capabilities requires retraining with the original data, which is both computationally demanding and time-consuming (Radford et al., 2021; Huyen, 2022; Parisi et al., 2019; Kirkpatrick et al., 2017).

Researchers have proposed several strategies to overcome catastrophic forgetting, with *replay* (or rehearsal) being a prominent approach. Replay involves mixing new samples with examples from previously learned tasks to reinforce older knowledge (Wilson & McNaughton, 1994; Rasch & Born, 2007; Shin et al., 2017; Rebuffi et al., 2017). Often, this approach is implemented through a memory buffer that stores and revisits past data. Other biologically inspired approaches, such as spiking neural networks leveraging spike-timing-dependent plasticity, have also been explored (Zenke et al., 2017), but neural networks still lag behind human adaptability in retaining and integrating knowledge over time. Moreover, in reinforcement learning, Prioritized Experience Replay (PER) highlights the idea of replaying more significant experiences more frequently (Schaul et al., 2015), focusing on buffer sampling based on temporal-difference (TD) errors rather than buffer allocation.

While replay buffers help mitigate forgetting, a significant challenge remains: the *limited buffer size* forces careful sampling of past data to ensure broad coverage (Rolnick et al., 2019; Parisi et al., 2019). Balancing the need to revisit earlier tasks with the need to incorporate new information quickly becomes increasingly complex when resources are limited. Thus, identifying the most impactful samples to store is crucial.

In this paper, we propose a novel data selection strategy for continual learning that focuses on "easy-to-forget" samples [1]. By prioritizing the data points most at risk of causing performance degradation if excluded, our method tries to more effectively mitigates catastrophic forgetting. We provide empirical results demonstrating improved retention of past knowledge and efficient adaptation to new tasks, offering a practical approach for real-world continual learning scenarios.

[1] `https://github.com/arahmoun2002/dl-forgetting`

2. Methodology

2.1. Our idea

Our approach prioritizes the retention of “easy-to-forget” samples in the experience replay buffer. The core challenge lies in both identifying these samples and ensuring that the overall data distribution remains well-represented. Focusing exclusively on harder-to-learn (and potentially outlier) samples risks losing coverage of the core distribution.

Previous work (Toneva et al., 2018) has shown that “unforgettable” examples tend to be learned early during training, implying that samples which are harder to learn (or learned later) are also the easiest to forget. Consequently, we design our replay buffer to emphasize samples that remain difficult for the model to predict, particularly during the later stages of training. This approach naturally retains a sufficient number of “easier” samples as well, striking a balance between “easy” and “hard” examples. Thus, the buffer captures both the representative portion of the data and the more extreme cases.

In this way, the model can frequently revisit the most challenging and informative samples, helping preserve knowledge from past tasks without overburdening the buffer. To maintain this balance, we introduce several hyperparameters that prevent challenging samples from disappearing too quickly while also keeping them from overwhelming the rest of the data.

2.2. Implementation

In this section, we describe how we implemented our method in practice. We present a pseudo-code 1 algorithm for better clarity on our buffer update algorithm, outlining each of the key steps to show how the method operates end-to-end within a training loop.

2.2.1. Weight Initialization, Accumulation, and Normalization

At the start of each task, we initialize an array of weights for every sample. Each weight begins at a constant *initial_weight* > 0 and is limited by a higher *cropping_weight* to keep the difference between correctly classified and misclassified samples within a reasonable range.

During training, we compute the probability of misclassification $P_{\text{miss}}(x_i)$ for each sample x_i in every batch. These probabilities are accumulated into the sample's weight across epochs. Specifically, after epoch t, the weight for sample i is updated as:

$$w_i^{(t)} = w_i^{(t-1)} + P_{\text{miss}}(x_i) \cdot g(t),$$

where $g(t) = t^k$ is a growth factor controlled by a hyperparameter k. This causes weights to increase more when misclassifications persist over multiple epochs, marking them as harder samples.

Once all epochs for a given task are completed, the accumulated weights are normalized to ensure they lie in $[0, 1]$ and remain comparable across samples (and across tasks). We introduce a *scale_factor* hyperparameter to regulate the impact of these misclassification counts. The final normalized weight for sample i is:

$$w_i^{\text{normalized}} = \frac{e^{(\text{scale\_factor} \cdot w_i)}}{\sum_{j=1}^{N} e^{(\text{scale\_factor} \cdot w_j)}},$$

where N is the total number of samples. This softmax-like normalization prevents samples from one task from dominating samples from another.

2.2.2. Buffer Update Using Multinomial Sampling

At the end of each task, we use the normalized weights to decide which samples to add to the buffer. We store both the sample and its weight. If adding a new sample causes the buffer to exceed its maximum size, we remove one based on a multinomial distribution, where the chance of removing sample i is proportional to $w_i^{\text{normalized}}$. As a result, samples with higher weights—those that are more challenging—are more likely to remain.

Because the buffer is updated constantly, older samples risk being replaced by newer ones. To address this, we introduce a parameter *weights_multiplier* > 1. After each update, we multiply the current weights in the buffer by this factor, ensuring that older samples do not vanish too quickly.

2.2.3. Training

Each training step starts by computing the classification loss on a batch from the current dataset. We then retrieve additional samples from the replay buffer, where the number of replay samples is set by multiplying the main batch size by the hyperparameter *rehearsal_ratio*.

A higher *rehearsal_ratio* places more emphasis on retaining past knowledge, but can slow adaptation to new data; a lower ratio focuses more on the current task at the risk of forgetting earlier tasks. We compute the loss on these buffer samples and add it to the main loss for backpropagation, ensuring the model balances learning from new data with retaining past experiences.

3. Results

Implementation Details

Model and Architectures. We evaluate the proposed method using a ResNet18 model for CIFAR10 and CI-

Algorithm 1 Proposed Buffer Update Algorithm with Weighted Sampling

Input: Dataset $\mathcal{D}$, Buffer $\mathcal{B}$, Maximum Buffer Size M, Epochs E, Initial Weights w_{init}, Cropping Weight w_{crop}, Growth Factor $g(t)$, Scale Factor λ, Weight Multiplier γ
Output: Updated Buffer $\mathcal{B}$

$\mathbf{W} \leftarrow [w_{\text{init}}, \ldots, w_{\text{init}}]$
for each task $\mathcal{T}$ **do**
  for epoch $t = 1$ to E **do**
    for each sample $x_i \in \mathcal{D}$ **do**
      $P_{\text{miss}}(x_i) \leftarrow 1 - P_{\text{correct}}(x_i)$
      $\mathbf{W}[i] \leftarrow \mathbf{W}[i] + P_{\text{miss}}(x_i) \cdot g(t)$
      $\mathbf{W}[i] \leftarrow \min(\mathbf{W}[i], w_{\text{crop}})$
    end for
  end for
  $\mathbf{W}_{\text{norm}}[i] \leftarrow \frac{\exp(\lambda \cdot \mathbf{W}[i])}{\sum_{j=1}^{|\mathcal{D}|} \exp(\lambda \cdot \mathbf{W}[j])}$
  for each sample $x_i \in \mathcal{D}$ **do**
    $\mathcal{B} \leftarrow \mathcal{B} + \{x_i, \mathbf{W}_{\text{norm}}[i]\}$
    if $|\mathcal{B}| > M$ **then**
      $j \sim$ Multinomial$(Weights(\mathcal{B}))$
      $\mathcal{B} \leftarrow \mathcal{B} \setminus \{x_j\}$
    end if
  end for
  for each sample $x_i \in \mathcal{B}$ **do**
    $\mathbf{W}[i] \leftarrow \min(\mathbf{W}[i] \cdot \gamma, w_{\text{crop}})$
  end for
end for

FAR100, and a fully connected network with two hidden layers, each containing 100 ReLU units, for MNIST. The ResNet18 model is modified and trained from scratch without pretraining, following the setup described in *Continual Learning with Strong Experience Replay (SER)* (Zhuo et al., 2023). These architectures are chosen for their balance between computational efficiency and representational power, making them suitable for continual learning scenarios.

CL Settings. The proposed method is evaluated in three continual learning (CL) settings: Class Incremental Learning (Class-IL), Task Incremental Learning (Task-IL), and Domain Incremental Learning (Domain-IL). In Class-IL, the model incrementally learns new classes with each task, and the task identity is not provided during inference. In Task-IL, the task identity is available during testing, allowing the model to focus on a specific subset of classes for each task, making this an easier setting. Domain-IL requires the model to adapt to shifts in the input distribution across tasks while retaining previously learned knowledge. These settings assess the method's ability to handle various challenges in continual learning, such as task ambiguity and domain shifts.

Training. We use the Stochastic Gradient Descent (SGD) optimizer for training all models. CIFAR10 is trained for 20 epochs per task, while CIFAR100 is trained for 50 epochs. MNIST is trained for one epoch per task due to its simplicity. A MultiStepLR scheduler is used to decay the learning rate by a factor of 0.1 at specific milestones, with epoch 15 for CIFAR10 and epochs 35 and 45 for CIFAR100. The batch size and initial learning rate are consistent with prior works for fair comparisons. Data augmentation strategies, including random cropping and horizontal flipping, are applied to both stream and buffer samples for CIFAR datasets, while no augmentation is applied to MNIST. We selected these parameters to ensure alignment with the baselines used in previous studies, specifically DER++ (Buzzega et al., 2020) and SER (Zhuo et al., 2023).

Hyperparameter Settings

For our implementation, we set the hyperparameters as follows: the initial weight for sample differences (`initial_weight`) was fixed at 1×10^{-3}, the scaling factor (`scale_factor`) at 0.1, the weight multiplier (`weight_multiplier`) at 1.2, and the rehearsal ratio (`rehearsal_ratio`) at 1.0. To prevent numerical instability, the maximum weight value (`cropping_weight`) was limited to 10^7, and the growth factor (`growth_factor`) was set to 1.1 to emphasize misclassified samples over time. While these parameters were carefully selected based on practical experimentation, we acknowledge that tuning hyperparameters in a continual learning framework is inherently challenging due to dynamic interactions between tasks. Thus, the chosen values may not represent the optimal configuration for all scenarios, leaving room for future exploration with automated or systematic search methods.

Metrics

We evaluate the continual learning methods with two metrics: **final average accuracy** and **average forgetting**. Let $a_{T,t}$ denote the testing accuracy on the t-th task when the model is trained on T-th task, and the final average accuracy A_t on all T tasks is computed as:

$$\text{Forgetting} = \frac{1}{T-1} \sum_{t=1}^{T-1} \max_{i \in \{1,\ldots,T-1\}} (a_{i,t} - a_{T,t})$$

4. Discussion

Our results show that the proposed method, EF (Easy-to-Forget sampling), helps reduce the problem of catastrophic forgetting in continual learning. This section discusses how our findings support the claims made in the introduction, compares our method with existing approaches, and highlights the strengths, weaknesses, and possible impacts of

Table 1. Classification results on benchmark datasets (CIFAR10, CIFAR100, P-MNIST, and R-MNIST) averaged across 5 runs. Results are reported for Class-IL, Task-IL, and Domain-IL settings, including results from EF (our method). The best-performing technique for each metric is highlighted in bold.

Buffer Size	Method	CIFAR10		CIFAR100		MNIST	
		Class-IL	Task-IL	Class-IL	Task-IL	P-MNIST Domain-IL	R-MNIST Domain-IL
-	Joint	**92.20 ± 0.15**	**98.31 ± 0.12**	**70.21 ± 0.15**	**85.25 ± 0.29**	**94.33 ± 0.17**	**95.76 ± 0.04**
	SGD	19.62 ± 0.05	61.02 ± 3.33	17.27 ± 0.14	42.24 ± 0.33	40.70 ± 2.33	67.66 ± 8.53
200	ER (Rolnick et al., 2019)	44.79 ± 1.86	91.19 ± 0.94	21.94 ± 0.83	62.41 ± 0.93	72.37 ± 0.87	85.01 ± 1.90
	DER++ (Buzzega et al., 2020)	64.88 ± 1.17	91.92 ± 0.60	27.46 ± 1.16	62.55 ± 2.31	**83.58 ± 0.59**	90.43 ± 1.87
	SER (Zhuo et al., 2023)	**69.86 ± 1.28**	**94.51 ± 0.61**	**47.96 ± 1.03**	**77.12 ± 0.34**	82.76 ± 0.62	**91.78 ± 0.81**
	GEM (Lopez-Paz & Ranzato, 2017)	25.54 ± 0.76	90.04 ± 0.94	19.73 ± 0.34	57.13 ± 0.94	66.93 ± 1.25	80.80 ± 1.15
	EF (Our Method)	63.43 ± 1.50	93.46 ± 1.90	32.85 ± 1.10	56.24 ± 0.85	67.01 ± 1.20	74.80 ± 0.90

Table 2. Average forgetting of ER and its expansions on CIFAR10 (5 tasks) and CIFAR100 (20 tasks). Lower is better. Buffer size is 200 for both datasets.

Method	CIFAR10		CIFAR100	
	Class-IL	Task-IL	Class-IL	Task-IL
ER(Rolnick et al., 2019)	54.12	5.72	83.54	23.72
DER++ (Buzzega et al., 2020)	39.84	8.25	71.02	20.64
SER (Zhuo et al., 2023)	**20.25**	**2.52**	**60.61**	**13.61**
EF (Our Method)	47.31	8.02	66.57	22.14

our idea.

Evidence Supporting Our Claims

Our method, EF (Easy-to-Forget sampling), helps reduce catastrophic forgetting, as shown in the results. For example, in CIFAR-10 Class-IL, EF achieves an accuracy of 63.43%, which is much better than ER (44.79%) and GEM (25.54%). Table 2 also shows that EF lowers forgetting rates. In CIFAR-10 Class-IL, EF's forgetting rate is 47.31%, while ER's is higher at 54.12%. These results support our idea that focusing on easy-to-forget samples helps the model remember better.

Baseline Comparison

EF works well in simpler datasets and with small buffer sizes. For example, in MNIST, EF achieves good results in Domain-IL and Task-IL settings, similar to advanced methods. However, EF struggles with more difficult datasets like CIFAR-100 Task-IL, where methods like SER perform better. SER uses a special way to keep outputs consistent between tasks, which helps it handle complex data. EF's weighted sampling is a good step forward, but it still needs improvement to handle harder tasks and reduce task interference.

Strengths and Weaknesses

One of EF's strengths is its simplicity and low computational requirements. By focusing on harder-to-learn samples, EF helps the model remember important information without requiring complex methods. This makes EF a good choice for systems with limited resources. However, EF has some weaknesses. It doesn't perform as well on more complex tasks, like CIFAR-100 Class-IL, where tasks interfere with each other. Additionally, EF doesn't handle class imbalances well, which can affect its performance on harder datasets. These limitations show that there is room to improve EF.

Implications and Future Work

EF could benefit various applications, including robotics and personalized systems, where retaining rare or challenging examples is crucial. Its efficiency makes it well-suited for resource-limited devices, such as mobile or edge platforms. In the future, refining the weighting mechanism to address class imbalances and task complexity could further boost EF's effectiveness. However, identifying the right hyperparameters remains a challenging task; more investigation is needed to fully optimize EF for different scenarios.

5. Summary

We introduced a data selection strategy that prioritizes "easy-to-forget" samples in the experience replay buffer, under the hypothesis that hard-to-learn examples are most likely to be forgotten. By assigning higher weights to samples that remain misclassified over several epochs and probabilistically including them in the buffer, our method aims to reinforce knowledge retention across tasks.

Empirical evaluations on CIFAR-10, CIFAR-100, and MNIST demonstrate that our approach preserves past knowledge more effectively than the less complex baselines, though it does not surpass more advanced techniques. Nonetheless, focusing on persistently misclassified samples shows potential for a balanced trade-off between computational cost and the need to address catastrophic forgetting, making it a practical option for resource-constrained continual learning.

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